

Satyam Goyal

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EDUCATION

University of Michigan, Ann Arbor

M.S.E in Computer Science Engineering; GPA: 3.95/4.0
B.S. in Computer Science

Ann Arbor, MI

Expected Aug. 2026

SELECTED PUBLICATIONS

Fairness Transformer: A Discrete Division Architecture | *NeuRIPS 2026 Pending*

ECHO: Sampling aware training for MDLMs | *NeuRIPS 2026 Pending*

Improving Sparse Memory Finetuning | *ICML 2026*

IOLBench: Benchmarking LLM Reasoning Capabilities

Encolor: Improving visual accessibility with a deep encoder-decoders | *AAAI 2024*

RESEARCH & ENGINEERING EXPERIENCE

University of Michigan AI Lab

Aug. 2024 – Present

Graduate Researcher

Ann Arbor, MI

Diffusion Language Models (ECHO) | *Advised by Prof. JJ Park*

- Developed ECHO, a sampling-aware MDLM training algorithm grounded in constrained Gibbs sampling (MCMC), outperforming similar SOTA methods by **15%** on reasoning benchmarks.
- Engineered a distributed training pipeline on 8xA100 clusters, scaling models to 50B tokens while reducing memory overhead by **30%**; pretrained and finetuned models of **100M to 8B** parameters.

Continual Learning & Alignment | *Advised by Prof. Samet Oymak*

- Retrofitted Qwen-2.5 with sparse memory layers to enable interpretable continual learning.
- Reduced catastrophic forgetting by **90%** w/ KL-divergence based finetuning algorithms.

Snowflake

May. 2026 – Present

AI Research Intern

Menlo Park, CA

- Built an adaptive platform that trained MTP speculators fine-tuned to customer needs, delivering up to **19% tok/s speedup** within **100M tokens** of training, leading to critical customer acquisitions.
- Member of core GLM 5.2 serving team (handling 100M tok/week), improved KV transfer speeds of **100k+ tokens to under 100ms**; developed routing layers in llm-d reducing **E2E latency by 17%**.
- Developed vLLM configs and kernel optimizers improving serving performance by **50%**, bringing Snowflake among the **top-3 DeepSeek V4 Pro** inference providers.
- Experimenting with **discrete diffusion models** to improve AR generation quality and speed.

IBM Research

May 2025 – Aug. 2025

AI Research Intern

Yorktown Heights, NY

- Reduced LLM serving costs and latency by **33%** by designing a KV-cache aware router for IBM, Google, and Red Hat's distributed LLM inference platform.
- Engineered an open-source vLLM simulator in Go and Python predicting TTFT and ITL with **95% request-level accuracy** at **500x** real-time speed.
- Authored comprehensive guides on vLLM internals, reaching 5,000+ developers to simplify accelerated inference.

INDEPENDENT RESEARCH & COLLABORATIONS

IOLBench - LLM Benchmarking | *In collab. with Dr. Soham Dan (Microsoft Research)* May 2024 – Oct. 2024

- Constructed a dataset of 1,000+ linguistic tasks to evaluate LLM reasoning capabilities.
- Revealed a **40% performance degradation** in logical reasoning across 10 SOTA models.
- Created a scalable Python evaluation library using NumPy and RegEx to standardize LLM testing methodologies.

SKILLS

Core ML & Systems: Discrete Diffusion Models, Alignment, Speculative Decoding, PD Disaggregation, KV Caching, Distributed Training, Continual Learning

ML/AI/Data: PyTorch, JAX, Hugging Face, Weights & Biases (W&B), Accelerate, Unsloth, Pandas

Cloud & DevOps: SLURM/HPC, vLLM, LM Cache, uv, Docker, Kubernetes, AWS

Languages: Python, Go, C++, SQL, JavaScript